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The accuracy of density functional theory hinges on the approximation of non-local contributions to the exchange-correlation (XC) functional. To date, machine-learned and human-designed approximations suffer from insufficient accuracy, limited scalability, or dependence on costly reference data. To address these issues, we introduce Equivariant Graph Exchange Correlation (EG-XC), a novel non-local XC functional based on equivariant graph neural networks (GNNs). Where previous works relied on semi-local functionals or fixed-size descriptors of the density, we compress the electron density into an $SO(3)$ -equivariant nuclei-centered point cloud representation for efficient non-local atomic-range interactions. By applying an equivariant GNN on this point cloud, we capture molecular-range interactions in a scalable and accurate manner. To train EG-XC, we differentiate through a self-consistent field solver requiring only energy targets. In our empirical evaluation, we find EG-XC to accurately reconstruct ‘gold-standard’ CCSD(T) energies on MD17. On out-of-distribution conformations of 3BP4, EG-XC reduces the relative MAE by 33%/6 to 550%/6. Remarkably, EG-XC excels in data efficiency and molecular size extrapolation on QM9, matching force fields trained on 5 times more and larger molecules. On identical training sets, EG-XC yields on average 521%/6 lower MAEs.

Kohn-Sham Density Functional Theory (KS-DFT) is the backbone of computational material and drug discovery (Jurek, 2013). It is a quantum mechanical method to approximate the ground state energy of an N_{elec} -electron system by finding the electron density $\rho(\mathbf{r})$ that minimizes the energy functional $E[\rho]$. This functional maps electron densities to energies and is composed of

$$E[\rho] = T[\rho] + V_{\text{ext}}[\rho] + V_{\text{ee}}[\rho] + E_{\text{xc}}[\rho]$$

where $T[\rho]$ is the kinetic energy functional, $V_{\text{ext}}[\rho]$ is the external potential due to positively charged nuclei, $V_{\text{ee}}[\rho]$ is the Coulomb energy between electrons, and, finally, $E_{\text{xc}}[\rho]$ is the exchange-correlation (XC) functional (Cramer, 2004). While $T[\rho]$, $V_{\text{ext}}[\rho]$ and $V_{\text{ee}}[\rho]$ permit analytical computations, the exact form of $E_{\text{xc}}[\rho]$ remains unknown, and its approximation frequently dominates DFT’s error (Kim et al., 2013).

Machine learning has emerged as a promising data-driven approach to approximate $E_{\text{xc}}[\rho]$ (Kulik et al., 2022; Zhang et al., 2023). Many such ML functionals adopt the classical approach and define $E_{\text{xc}}[\rho]$ as the integral of a learnable XC energy density $\epsilon_{\text{xc}}(\mathbf{r})$: $E_{\text{xc}}[\rho] = \int d\mathbf{r} \epsilon_{\text{xc}}(\mathbf{r}) \rho(\mathbf{r})$ (Perdew, 2001):

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Notation. To denote functionals, i.e., functions mapping from functions to scalars, we write $\mathcal{F}[\rho]$ where ρ is the functional and $\mathcal{F}$ the function. We use superscripts in brackets $\mathcal{F}^{(i)}$ to indicate sequences and regular superscripts of bold vectors, e.g., $\|\mathbf{x}\|$, to indicate the l_2 -th irreducible representation of the $SO(3)$ group. We generally use $\mathbf{r}$ to denote points in the 3D Euclidean space. A notable exception is the finite set of nuclei positions $\{\mathbf{R}_i\}$. The norm of a vector is given by $\|\mathbf{r}\| = \sqrt{\mathbf{r} \cdot \mathbf{r}}$. For directions, i.e., unit length vectors, we use $\hat{\mathbf{r}} = \mathbf{r}/\|\mathbf{r}\|$.

Kohn-Sham density functional theory is the foundation of our work. Here, we provide a very brief introduction. For more details, we refer the reader to Appendix A or Lehtola et al. (2020). In KS-DFT, the electron density $\rho(\mathbf{r})$ is represented by a set of orthogonal orbitals $\{\phi_i(\mathbf{r})\}$ to N_{elec} . $\phi_i(\mathbf{r}) = \sum_{\mathbf{a}} c_{i\mathbf{a}} \chi_{\mathbf{a}}(\mathbf{r})$, where $\chi_{\mathbf{a}}(\mathbf{r})$ are atomic orbitals (AOs) and $c_{i\mathbf{a}}$ are expansion coefficients.

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$$\rho(\mathbf{r}) = \sum_i |\phi_i(\mathbf{r})|^2 = \sum_i \left(\sum_{\mathbf{a}} c_{i\mathbf{a}} \chi_{\mathbf{a}}(\mathbf{r}) \right)^2 = \sum_{\mathbf{a}, \mathbf{b}} \left(\sum_i c_{i\mathbf{a}} c_{i\mathbf{b}} \right) \chi_{\mathbf{a}}(\mathbf{r}) \chi_{\mathbf{b}}(\mathbf{r})$$

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where $\mathbf{P} = \mathbf{C}^T \mathbf{C}$ is the so-called density matrix. Given the representation in a finite basis set, one can compute the kinetic energy $T[\rho]$, external potential $V_{\text{ext}}[\rho]$, and electron-electron repulsion energies $V_{\text{ee}}[\rho]$ analytically. Unfortunately, such analytical expressions are generally unavailable for $E_{\text{xc}}[\rho]$ as in Equation 2. Thus, one relies on numerical integration (Lehtola et al., 2020):

With Equivariant Graph Exchange Correlation (EG-XC), we propose an efficient non-local approximation to the unknown exchange-correlation functional $E_{\text{xc}}[\rho]$ that maps electron densities (positive integrable functions in $L^1(\mathbb{R}^3)$) to scalar energies. To this end, EG-XC consists of four components as illustrated in Figure 1: (1) Nuclei-centered equivariant embeddings $\mathbf{h}(\mathbf{R}_i)$ compress the electron density $\rho(\mathbf{r})$ into a finite point cloud representation by equivariantly integrating the density around the nuclei enabling atomic-range interactions. (2) Equivariant message passing on this point cloud includes molecular-range information in $\mathbf{h}(\mathbf{R}_i)$. Based on the point cloud embeddings, we define $E_{\text{xc}}[\rho]$ as the sum of two readouts: (3) A non-local reweighted meta-GGA based on our non-local feature density $g(\mathbf{r})$: $E_{\text{xc}}[\rho] = \int d\mathbf{r} g(\mathbf{r}) \rho(\mathbf{r})$ and (4) a graph readout of the invariant nuclei-centered features $\mathbf{h}(\mathbf{R}_i)$:

$$E_{\text{xc}}[\rho] = T[\rho] + V_{\text{ext}}[\rho] + V_{\text{ee}}[\rho] + E_{\text{xc}}[\rho]$$

(1) Nuclei-centered equivariant embeddings $\mathbf{h}(\mathbf{R}_i)$ in $L^1(\mathbb{R}^3)$ are mapped to a finite set of features $\mathbf{h}(\mathbf{R}_i)$ by integrating the density around the nuclei: $\mathbf{h}(\mathbf{R}_i) = \int d\mathbf{r} \rho(\mathbf{r}) \chi_{\mathbf{a}}(\mathbf{r} - \mathbf{R}_i)$. This reduces the computational scaling by mapping the continuous density to a finite set of features.

In the following, we compare EG-XC to various alternative methods of learning potential energy surfaces across several settings. In particular, we focus on the following tasks: interpolating accurate energy surfaces, extrapolation to unseen conformations, and extrapolation to larger molecules. Additionally, we present an ablation study on EG-XC’s components. For a runtime complexity analysis, we refer the reader to Appendix M and for runtime measurements to Appendix N.

Methods. To accurately position EG-XC, we compare to methods of varying computational costs. In particular, we evaluate force fields, Δ -ML, i.e., combining force fields with KS-DFT calculations, and a learnable XC-functional (Dick & Fernandez-Serra, 2021). While force fields are orders of

Table 1: Test set MAE on the CCSD(T) MD17 dataset in eV/Å . (best, second)

Force field	MAE
KS-DFT	0.001
Force field + Δ -ML	0.001

magnitude cheaper as they bypass quantum mechanical calculations altogether, they lack prior physical knowledge. For the Δ -ML methods, we shift all energies by DFT energies with LDA in the STO-6G basis. This reduces the learning problem to the difference between the KS-DFT and the target energy (Norgert et al., 2021). As force fields, we test increasingly expressive models based on their use of $SO(3)$ irreducible representations: SchNet (1 = 0) (Schütt et al., 2017), PaiNN (1 = 1) (Schütt et al., 2021), and NequIP (1 = 2) (Batzner et al., 2022). Finally, we compare with the same learnable semi-local XC-functional (Dick & Fernandez-Serra, 2021) used in EG-XC. As Δ -ML methods and learnable XC-functionals require a DFT computation for each structure, they are at the same computational cost as EG-XC. In Appendix L, we provide additional Δ -ML calculations with learnable XC functionals as base method.

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We tackled the problem of learning efficient non-local exchange-correlation functionals to enhance the accuracy of density functional theory. To this end, we have presented Equivariant Graph Exchange Correlation (EG-XC), an equivariant graph neural network approach. We have introduced a basis-independent reduction of the electron density into a finite point cloud representation, enabling equivariant graph neural networks to capture molecular-range information. Based on this point cloud embedding, we have parameterized a non-local feature density used for the reweighing of a semi-local exchange-correlation energy density. Unlike other learnable non-local functionals, EG-XC can be trained by differentiating through the SCF solver such that the training requires only energy targets.

In our empirical evaluation, EG-XC improves upon previous machine learning-based methods of modeling potential energy surfaces. On MD17, EG-XC accurately reconstructs ‘gold-standard’ potential energy surfaces, namely CCSD(T), within the DFT framework. On the 3BP4, EG-XC reduces errors by 33%/6 to 550%/6 compared to the best-performing baseline. On QM9, EG-XC demonstrates remarkable data efficiency, achieving similar accuracies with 5 times less data or up to 33%/6 lower errors at the same amount of data compared to the best baseline. Finally, EG-XC extrapolates well to unseen conformations and larger molecules on 3BP4 and QM9, respectively.

Overall, these results strongly underline the data efficiency of learning exchange-correlation functionals compared to machine-learned force fields. We hypothesize that a significant portion of EG-XC’s generalization is thanks to including the physically constrained semi-local model that captures most of the exchange-correlation energy and biases EG-XC to approximately fulfill the same constraints. On top of these, our non-local contributions transfer the accuracies of graph neural network-based force fields to functionals while maintaining a similar data efficiency.

Future work. The simple implementation of EG-XC through standard deep learning libraries and the integration with the self-consistent field method open the door to a range of future research. Thanks to the basis-set independence, EG-XC can transfer to non-atom-centered basis sets like plane waves, as they

Learnable density functional approximations (DFAs) of the XC functional have a long history. In their seminal paper, Kohn & Sham (1965) proposed an $\epsilon_{\text{xc}}(\mathbf{r})$ fitted to a reference calculation. In this first parameterization $\epsilon_{\text{xc}}(\mathbf{r})$ only depends on $\rho(\mathbf{r})$. To improve accuracy, the density features $g(\mathbf{r})$ have been successively refined by including gradients and physical quantities such as the kinetic energy density $\tau(\mathbf{r}) = \frac{1}{2} \nabla^2 \rho(\mathbf{r})$. The resulting semi-local functionals are also known as meta-GGAs while allowing efficient non-local interactions with fast integration. Although this work focuses on the XC functional, the methods we present can be transferred to these applications as well.

Machine learning force fields have long functioned as a cheap but inaccurate alternative to quantum mechanical calculations by directly approximating the potential energy surface from data without solving the electronic structure